

Lecture Notes 6

The Short Run

Intermediate Macroeconomics, EC2211

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Contents and literature

- Economic fluctuations ("business cycles")
- The IS curve
- The central bank policy rate
- The Philips curve
- Using the IS-MP-PC framework

Literature:

- Jones (2024), ch. 8-12
- Buiter and Rahbari (2015), "What is a (global) recession?"
- Cerrato and Gitti (2023), "Inflation since COVID: supply versus demand"

What have we done so far and what is next?

So far:

- The long run: economic growth
- Microfoundations of labor supply, consumption and investment

Now the shorter run:

- The IS-MP-Phillips curve framework
- The AS-AD model
- Topics in monetary and fiscal policy
- Exchange rates, capital flows and the international monetary system

Long run versus short run

Classical economics:

- Emphasizes self-equilibrating forces
- In the long run, these forces bring about good economic outcomes

John Maynard Keynes (1923, "A tract on monetary reform")

- "But this long run is a misleading guide to current affairs. In the long run we are all dead."
- "Economists set themselves too easy, too useless a task if in tempestuous seasons they can only tell us that when the storm is long past the ocean is flat again."

Read more [about Keynes on CORE Econ](#)

Long run versus short run

Keynesian macroeconomics:

- Born as a separate sub-discipline after the 1930's Great Depression
- Obvious that long-run forces were not working well
- Increased focus on the workings of the economy in the short run
 - Adjustment to shocks
 - Active policy measures to stabilize the economy

Economic fluctuations

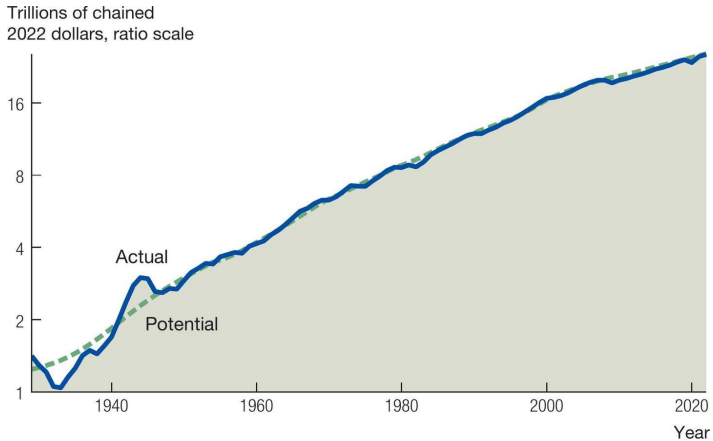
The short run

- Takes the long-run variables, potential output $\bar{Y}$ etc, as given
- Focuses on fluctuations around the long-run outcomes
- The **output gap** is the percent deviation of output from potential:¹

$$\tilde{Y}_t = \frac{Y_t - \bar{Y}_t}{\bar{Y}_t}$$

¹Jones uses the term "short-run output" for the output gap. I think that term is a bit misleading, and I have never heard anyone else using it. "Output gap" is the standard term for this concept.

U.S. real GDP: actual and potential



Source: U.S. Department of Commerce. Potential output is constructed by the Congressional Budget Office for years after 1949.

Figure 9.2 in Jones

Economic fluctuations

Often referred to as "business cycles"

- But fluctuations are not cyclical or regular
- Asymmetry: sharp contraction in economic activity often followed by gradual recovery

Expansion → contraction, boom → bust

Economic fluctuations and short-run output

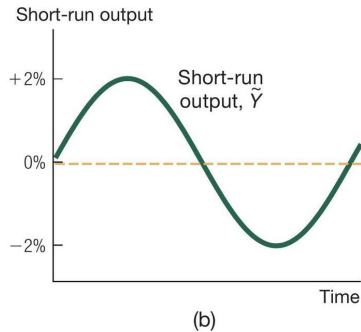
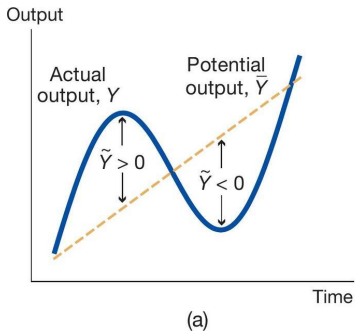
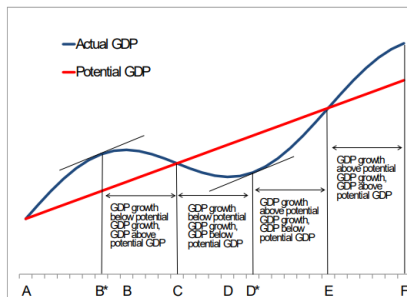


Figure 9.1 in Jones

What is a "recession" ?

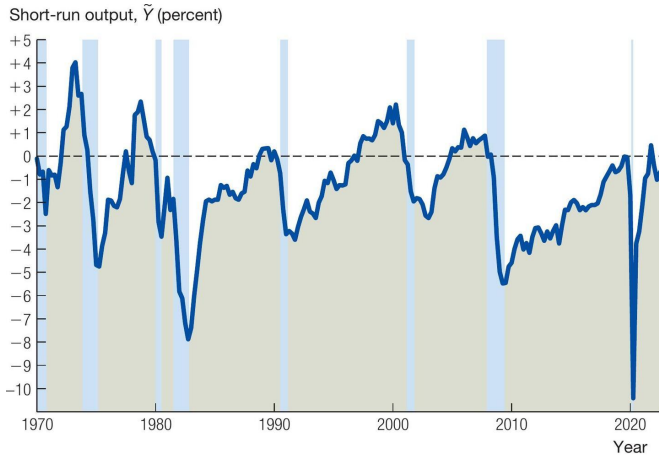
- When is the economy in recession?
- There is no formal definition
 - Negative output gap (C to E)
 - Negative output growth (B to D)
 - Output gap decreasing (B* to D*)
 - ...

Figure 7. Stylised Business Cycle with Positive Potential Growth and Peak-to-trough Criterion



From Buiter & Rahbari (2015)

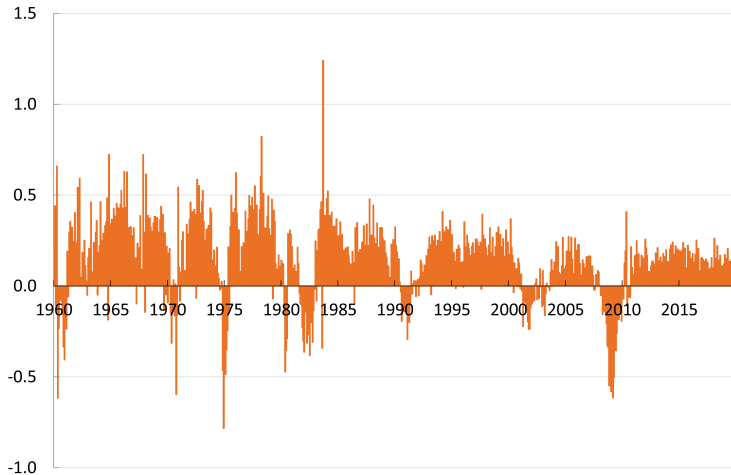
U.S. economic fluctuations and NBER recessions



Source: Federal Reserve Economic Data (FRED), courtesy of the Federal Reserve Bank of St. Louis. Potential output is constructed by the Congressional Budget Office. Periods of recession are shaded, using the dates assigned by the National Bureau of Economic Research.

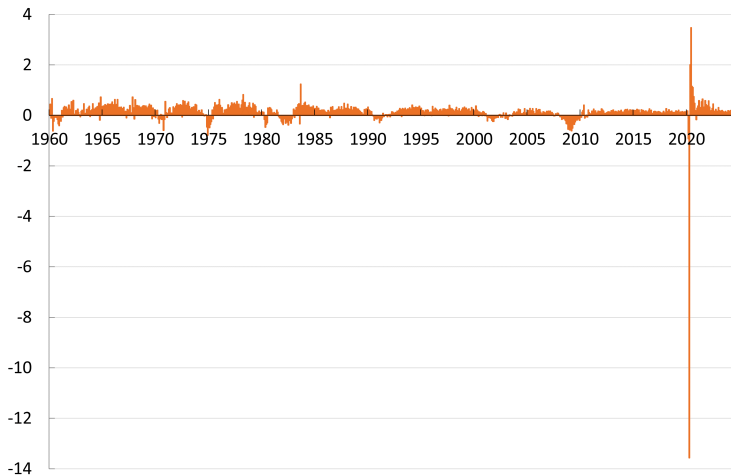
Figure 9.3 in Jones

Changes in U.S. employment



Percent monthly change. Source: FRED, total nonfarm employment

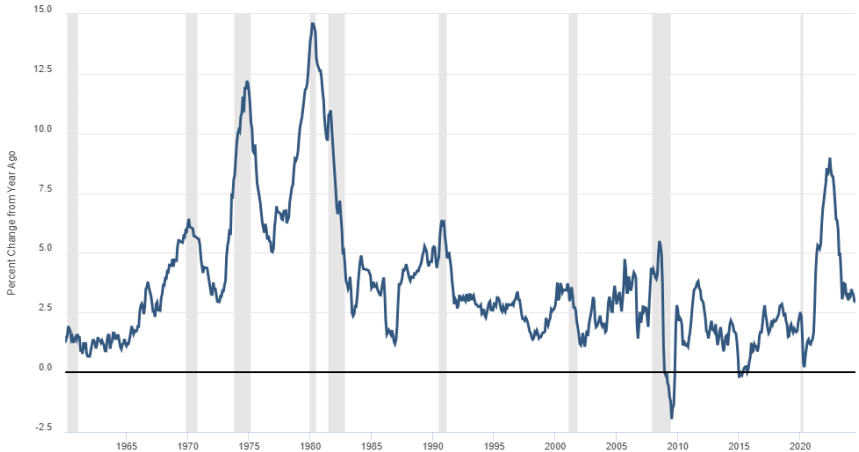
Changes in U.S. employment (now with 2020s!)



Percent monthly change. Source: FRED, total nonfarm employment

Inflation in the U.S.

FRED 



Shaded areas indicate U.S. recessions.

Source: U.S. Bureau of Labor Statistics

fred.stlouisfed.org

CPI inflation.

Economic fluctuations

Some episodes that I may refer to:

The Great Depression

1929-1930's deep global recession

The Great Inflation

1970's global episode with high inflation, slow growth

Volcker disinflation

Early 1980's, short but sharp recession

1990s crisis

Early 1990's, mostly Europe and related to fixed exchange rates

Global Financial Crisis, GFC

2007-2009, sometimes referred to as the *Great Recession*

Euro Crisis

ca 2009-2013, also referred to as the *European Debt Crisis*

Covid pandemic

2020-2021

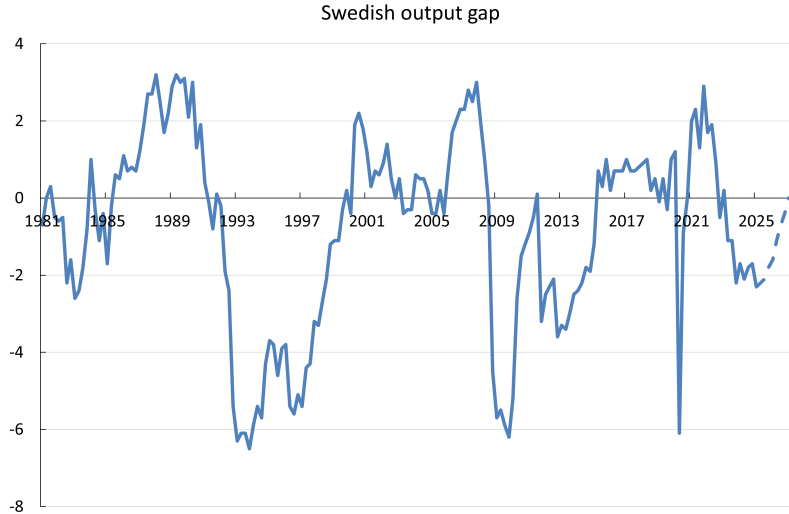
How deep is the current Swedish downturn?

The current recession is as severe as the one during the GFC

—Swedish minister of finance, September 20025

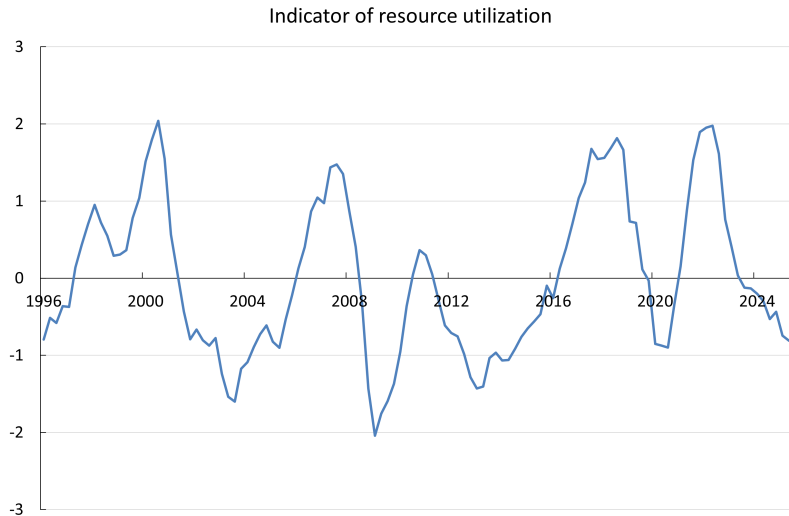
Do we agree?

Output gap according to Konjunkturinstitutet



Percent of potential GDP. Source: Konjunkturinstitutet, June 2025

Resource utilization according to Riksbanken



Standardized value. Source: Riksbanken, September 2025

The IS-MP-PC framework

Building the IS-MP-PC framework

The IS-MP-PC framework:

- A framework to analyze short-run economic fluctuations
- IS-MP: Interaction between household demand, fiscal policy, monetary policy and economic activity with a fixed price level
- PC: Phillips curve that shows the interaction between economic activity and changes in the price level (inflation)
- Analyze effects of shocks and monetary and fiscal policy

Building the IS-MP-PC framework

Three key components in the IS-MP-PC framework:

- An **IS curve** specifying how the output gap, $\tilde{Y}_t$, depends on the real interest rate, r_t ²
- A central bank that chooses a **policy rate**, i_t , with direct implications for the real interest rate r_t
- A **Phillips curve** relating output, $\tilde{Y}_t$, to inflation, π_t

In these notes, we will focus on the IS curve and the Phillips curve together with an interest rate set by the central bank – the MP curve

²I will follow the most common notation and let r denote the real interest rate, although Jones uses R

Building the IS-MP-PC framework

- The framework takes some shortcuts – it is not built from microfoundations
- We will first develop a basic version of this framework, skipping some interesting/important material in chapters 9-12
- We will eventually get back to some of these issues

Interest rates: concepts and notation

Jones: R = real interest rate, $\bar{r}$ = MPK

My notes: r = real interest rate, $r^* = \bar{r}$

Several concepts with somewhat varying definitions:

- **natural rate of interest**
- **neutral interest rate**
- **long-run equilibrium interest rate**
- **r star**
- ...

Think of all these concepts as being the same, determined by our long-run model of the economy

Knut Wicksell and the natural rate

- Wicksell (1898), "Geldzins und Guterpreise" (*Interest and Prices*)
- Introduced the concept of a **natural rate of interest**. Now sometimes referred to as the Wicksellian interest rate.



Knut Wicksell 1851-1926

The IS curve

The IS curve

Recall the national income identity:

$$Y_t = C_t + I_t + G_t + EX_t - IM_t \quad (1)$$

Assume:

- I_t depends negatively on the interest rate: $I_t = I(\bar{r}_t)$
- C_t , G_t , EX_t and IM_t are proportional to potential output

Then $\tilde{Y}_t$, the deviation of output from potential, depends negatively on the interest rate

The IS curve

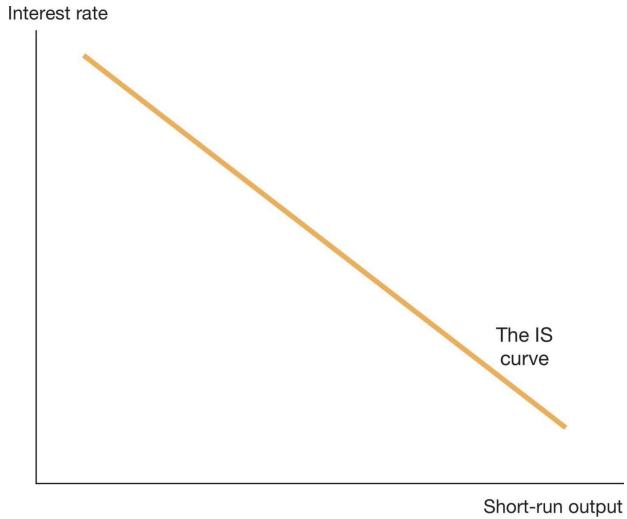


Figure 11.1 in Jones

Deriving the IS curve

But let us now follow Jones to derive the IS curve:

$$C_t = \alpha_C \bar{Y}_t \quad (2)$$

$$G_t = \alpha_G \bar{Y}_t \quad (3)$$

$$EX_t = \alpha_{EX} \bar{Y}_t \quad (4)$$

$$IM_t = \alpha_{IM} \bar{Y}_t \quad (5)$$

These assumptions are more "classical" than "Keynesian"

We will later on analyze more Keynesian versions

Deriving the IS curve

Investment as a share of potential output is:

$$\frac{I_t}{\bar{Y}_t} = \alpha_I - \beta(r_t - \bar{r}) \quad (6)$$

where you can think of $\bar{r}_t$ as the long-run average real interest rate determined by the marginal product of capital

Notation

Y_t	Output (GDP, production) at time t
$\bar{Y}_t$	Potential output
$\tilde{Y}_t$	Output gap ("short-run output"), $\tilde{Y}_t = (Y_t - \bar{Y}_t)/\bar{Y}_t$
C_t	Consumption
G_t	Government consumption
EX_t	Exports
IM_t	Imports
r_t	Real interest rate
$\bar{r}$	Natural rate of interest (MPK or r^*)
α_C	Consumption share of potential output
α_G	Government consumption share of potential output
α_{EX}	Export share of potential output
α_{IM}	Import share of potential output
α_I	Import share of potential output when $r_t = \bar{r}$
β	Parameter indicating how investment responds to interest rates

Deriving the IS curve

Use (2)-(6) in the national income accounts identity (1):

$$Y_t = \underbrace{\alpha_C \bar{Y}_t}_{C_t} + \underbrace{\alpha_G \bar{Y}_t}_{G_t} + \underbrace{[\alpha_I - \beta(r_t - \bar{r})] \bar{Y}_t}_{I_t} + \underbrace{\alpha_{EX} \bar{Y}_t}_{EX_t} - \underbrace{\alpha_{IM} \bar{Y}_t}_{IM_t}$$

and then divide by $\bar{Y}_t$:

$$\frac{Y_t}{\bar{Y}_t} = \alpha_C + \alpha_G + \alpha_I - \beta(r_t - \bar{r}) + \alpha_{EX} - \alpha_{IM} \quad (7)$$

Deriving the IS curve

We are interested in $\tilde{Y}_t$, the relative deviation of Y_t from $\bar{Y}_t$

Note that:

$$\tilde{Y}_t \equiv \frac{Y_t - \bar{Y}_t}{\bar{Y}_t} = \frac{Y_t}{\bar{Y}_t} - 1 \quad (8)$$

Let

$$\alpha \equiv \alpha_C + \alpha_G + \alpha_I + \alpha_{EX} - \alpha_{IM} - 1$$

and use with (8) in (7) to get **the IS curve**:

$$\tilde{Y}_t = \alpha - \beta(r_t - \bar{r}) \quad (9)$$

The IS curve

- In the long run, output is at its potential level ($Y_t = \bar{Y}_t$) and the interest rate is at its natural level ($r_t = \bar{r}$)
- Then $\tilde{Y}_t = 0$ and $\alpha = 0$
- Note that $\alpha = 0$ is just stating that the sum of expenditure shares sum to unity
- We will however consider temporary deviations in expenditure shares:
demand shocks

When r_t increases, we move up along the IS curve

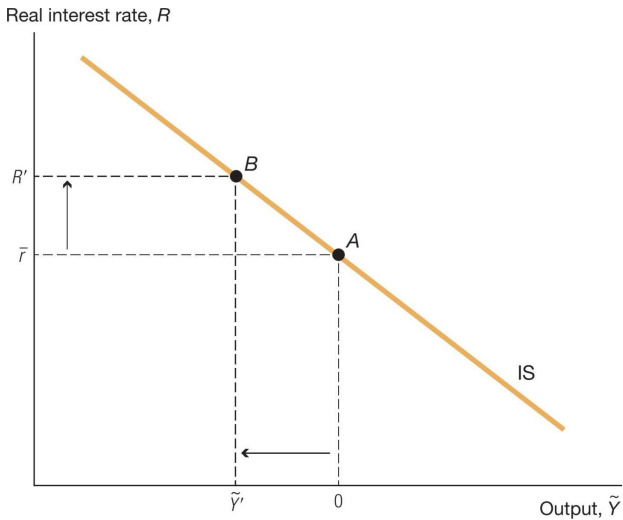


Figure 11.3 in Jones

When α increases, the IS curve shifts out

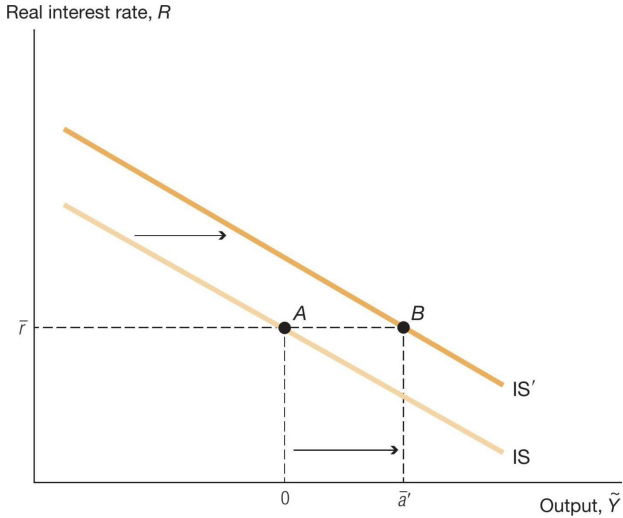


Figure 11.4 in Jones

Monetary policy

Policy rates

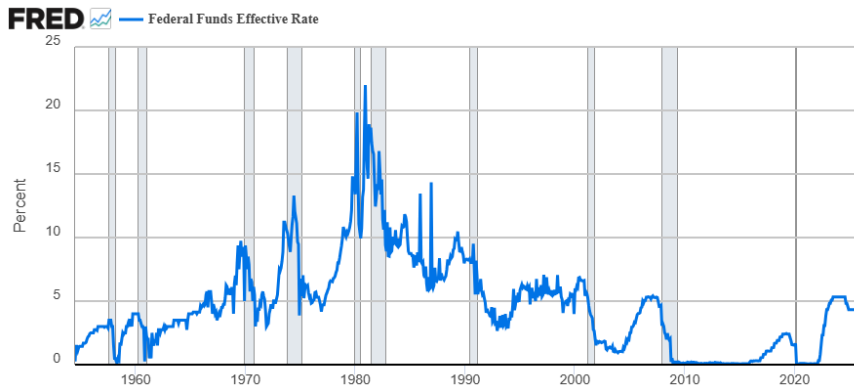
How is the real interest rate r_t determined?

The central bank sets a short-term nominal interest rate (i_t), its **policy rate**:

Fed	the federal funds target rate (currently a range)
ECB	key interest rates – deposit facility, main refinancing, marginal lending
Bank of England	Bank Rate
Riksbank	policy rate

We will get back to more details on what central banks do later

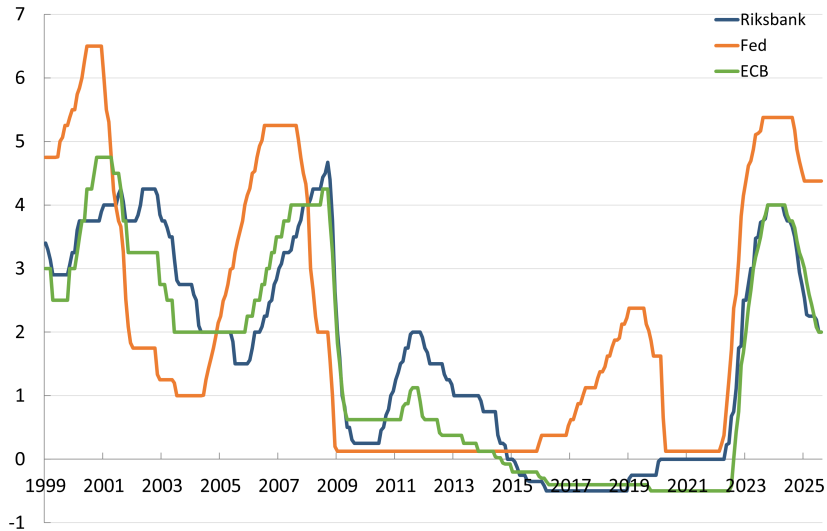
Federal funds rate



Source: Board of Governors of the Federal Reserve System (US) via FRED®
Shaded areas indicate U.S. recessions.

fred.stlouisfed.org

Policy rates



Nominal and real interest rates: the Fisher equation

The *ex ante* real interest rate equals the nominal interest rate minus expected inflation:

$$r_t = i_t - \pi_t^e$$

This is the expected return, adjusted for purchasing power of money, if investing at interest rate i_t

The MP curve assumes that expected inflation equals past inflation:

$$\pi_t^e = \pi_{t-1}$$

and that prices are sticky

The central bank can then control r_t by setting the nominal rate i_t

The IS-MP diagram

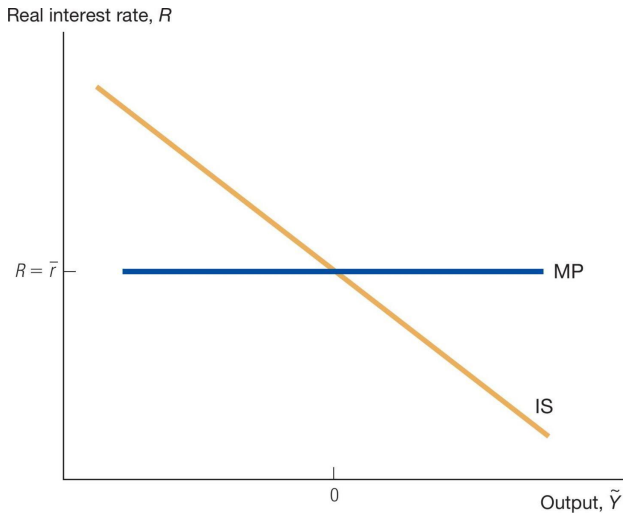


Figure 12.3 in Jones

An increase in $r_t \rightarrow$ MP curve shifts up, move along IS

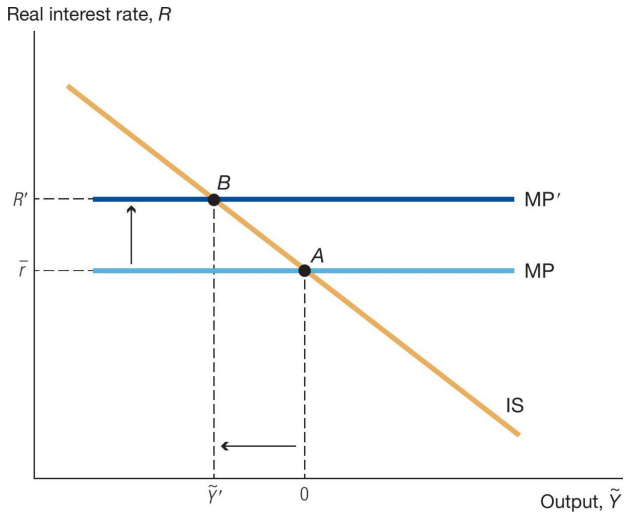


Figure 12.4 in Jones

If demand falls, the central bank can bring output back to potential by cutting the interest rate

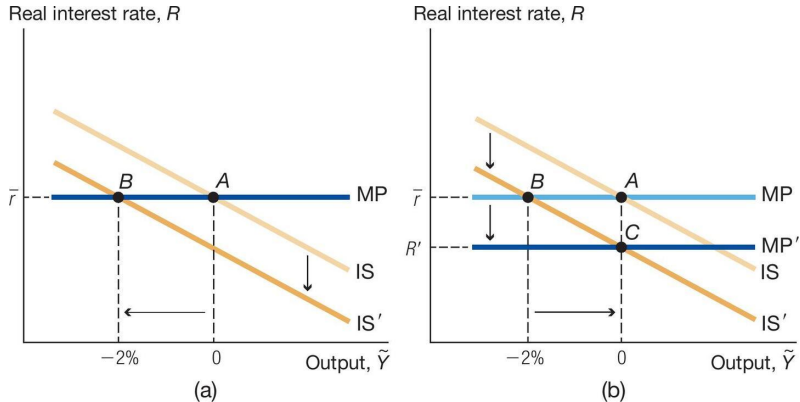


Figure 12.5 in Jones

The Phillips curve

The Phillips curve

- Named after A. W. Phillips who, in 1958, demonstrated a *negative relationship between unemployment and wage increases* in U.K. data from 1861-1913
- A similar negative relationship between unemployment and price increases (inflation) was then demonstrated for other economies
- Today, the Phillips curve is often stated as a *positive relationship between economic activity (e.g. the output gap) and expected inflation*

The original Phillips curve

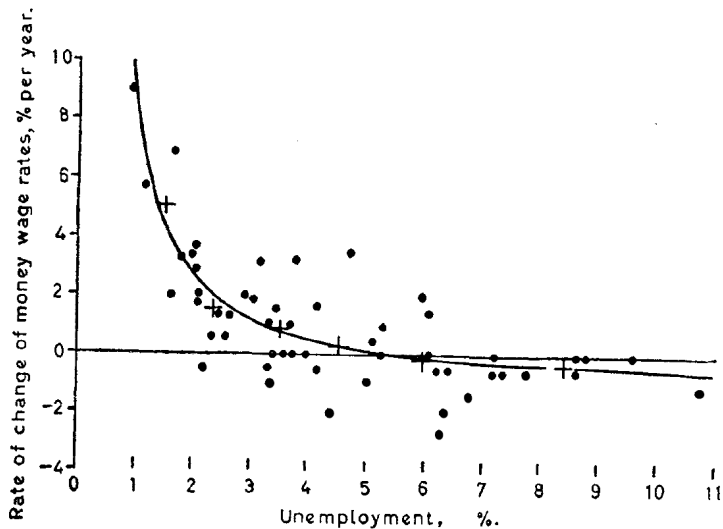


Fig.1.1861 – 1913

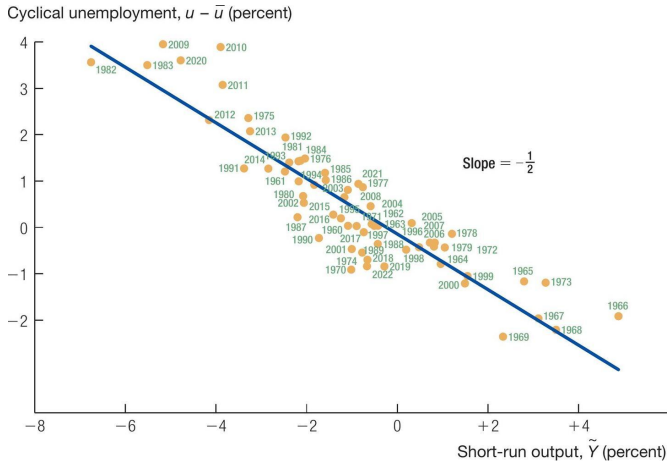
Okun's law

- Intuitively, we would expect unemployment to be high when economic activity is low and vice versa
- This also turns out to hold empirically
- In U.S. data, an increase in the output gap by two percentage points typically reduces the unemployment rate by one percentage points

$$u - \bar{u} \approx -\frac{1}{2} \tilde{Y}$$

- This relationship is named after Arthur Okun, 1960s

Okun's law



Source: FRED. U.S. Department of Commerce.

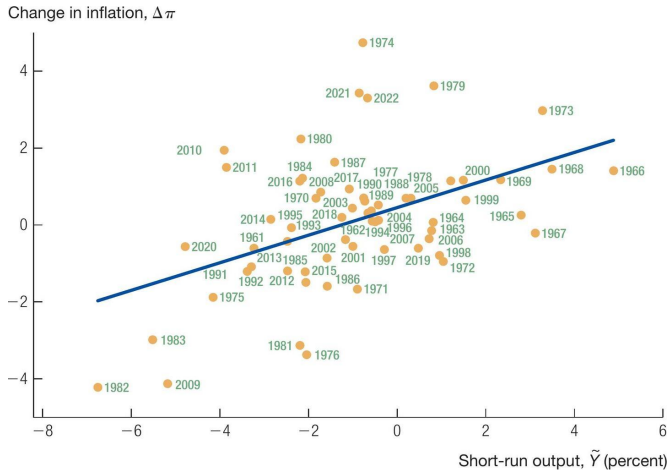
Figure 9.8 in Jones

The Phillips curve

Because of the Okun relationship, the Phillips curve can be formulated either as:

- a negative relationship between unemployment and (the change in) inflation
- or
- a positive relationship between the output gap and (the change in) inflation

The U.S. Phillips curve



Source: FRED. U.S. Department of Commerce.

Figure 9.7 in Jones

The Phillips curve and expectations

Can we reduce unemployment, raise economic activity, by allowing higher inflation?

- No, not in the long run!
- The relationship between inflation and activity only holds as long as inflation expectations are fixed
- The Phillips curve cannot be exploited – an insight anticipated by Edmund Phelps and Milton Friedman in the late 1960s
- ... which became evident in 1970s

The Phillips curve

$$\pi_t = \pi_t^e + \nu \tilde{Y}_t$$

The Phillips curve with adaptive expectations

We now assume that inflation expectations are *adaptive*:

$$\pi_t^e = \pi_{t-1}$$

The Phillips curve can then be rewritten as:

$$\pi_t = \pi_{t-1} + \nu \tilde{Y}_t + \sigma_t$$

where we also allow for an (exogenous) price shock σ (we may also refer to this shock as a "supply shock" or a "cost-push shock")

The Phillips curve

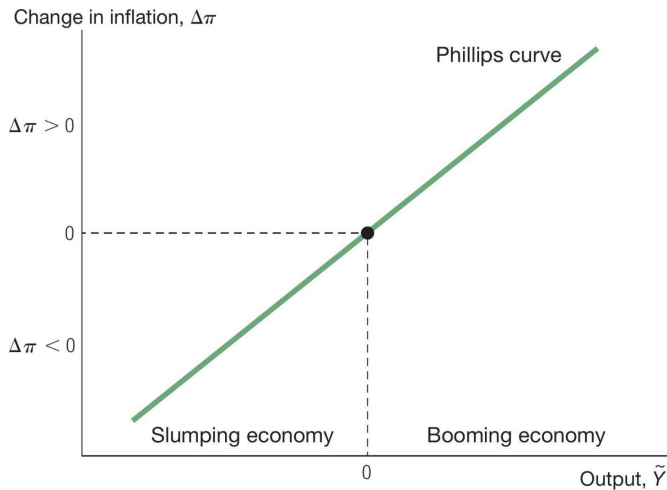


Figure 12.7 in Jones

Using the IS-MP-PC framework

The transmission mechanism of monetary policy

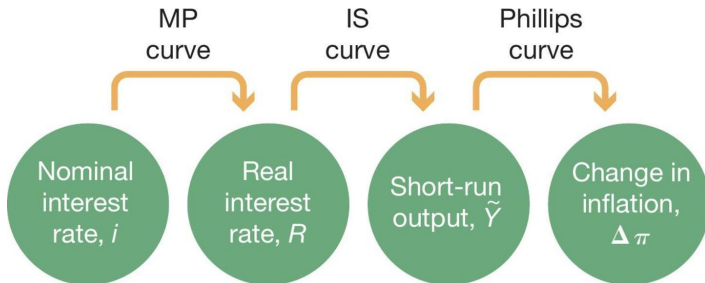
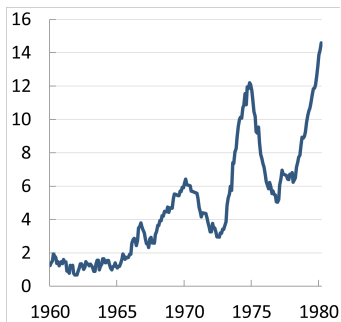


Figure 12.1 in Jones

The Volcker disinflation

U.S. inflation

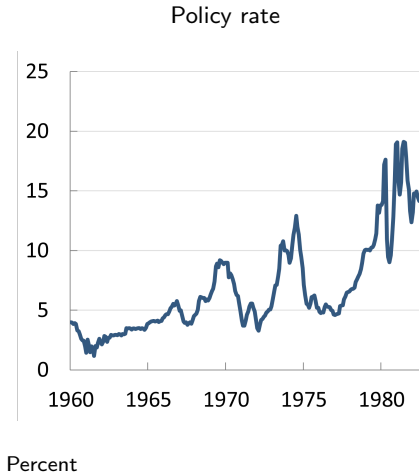


CPI, percent

- The economy has been hit by positive price shocks
- Phillips curve has shifted up; inflation is "too high", output is at potential level
- How can we make the economy return to an equilibrium with low inflation?

The Volcker disinflation

- Paul Volcker, Fed chairman from Aug. 1979
- Policy rate must be raised enough to generate a recession
- ... it was 10.5 percent when he became chairman
- ... raised to 19.1 percent in 1981



The Volcker disinflation

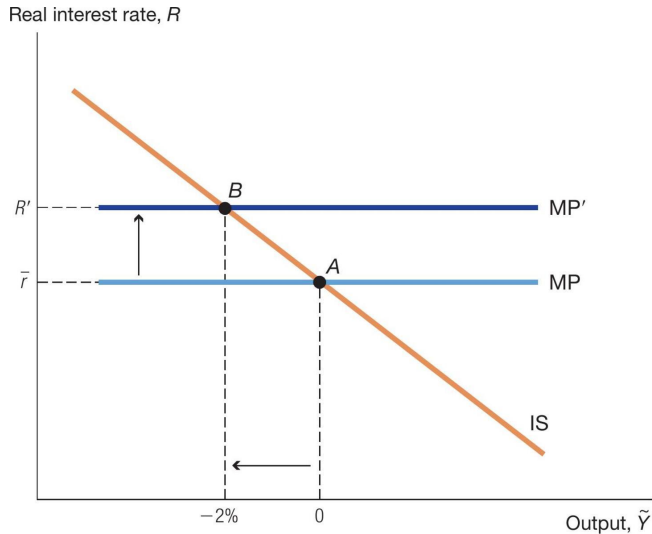


Figure 12.10 in Jones

The Volcker disinflation

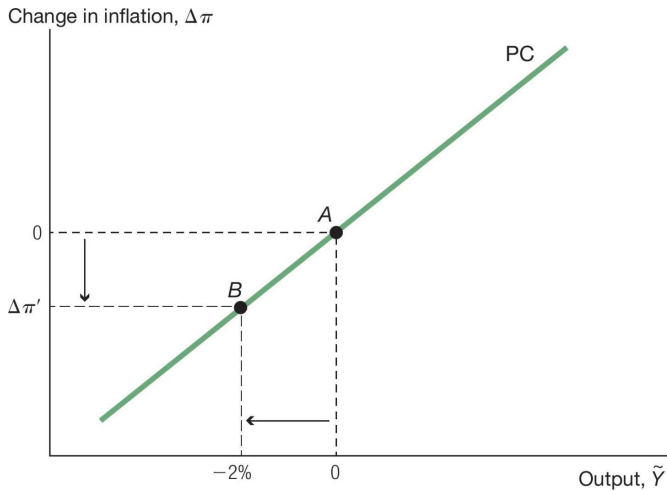


Figure 12.11 in Jones

The Volcker disinflation

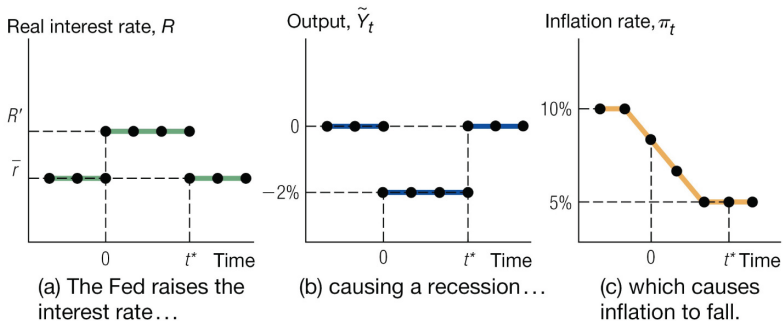
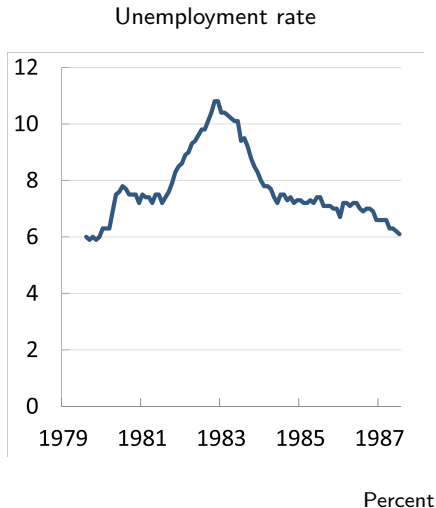


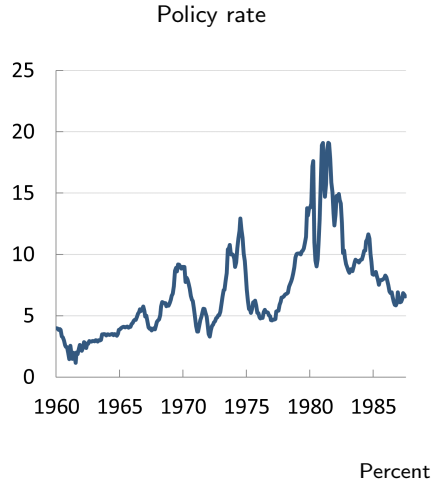
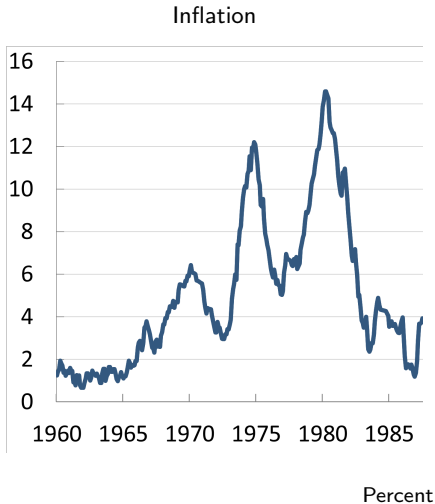
Figure 12.12 in Jones

The Volcker disinflation

- Unemployment 6 percent when Volcker became Fed chairman
- Rose to 10.8 percent in late 1982



The Volcker disinflation - inflation fell quickly and the policy rate could be reduced substantially



Navigating by the stars

- According to this framework: **Set $r < \bar{r}$ if $Y < \bar{Y}$** etc
- Easy?
- The main problem is to understand what $\bar{r}$ and $\bar{Y}$ are!
- Fed chairman Jay Powell has often said that monetary policy is like "navigating by the stars under cloudy skies"
(see e.g. [this speech](#) and recall that our $\bar{r}$ and $\bar{Y}$ are often referred to as r^* and y^*)

The great inflation of the 1970s

The **output gap**:

- A theoretical (but useful!) concept
- Not something that we can observe or measure
- Must be estimated/assessed
 - Estimates typically rely on historical trends
 - Deviations from "normal" development
 - Difficult to detect when the trend changes!

The great inflation of the 1970s

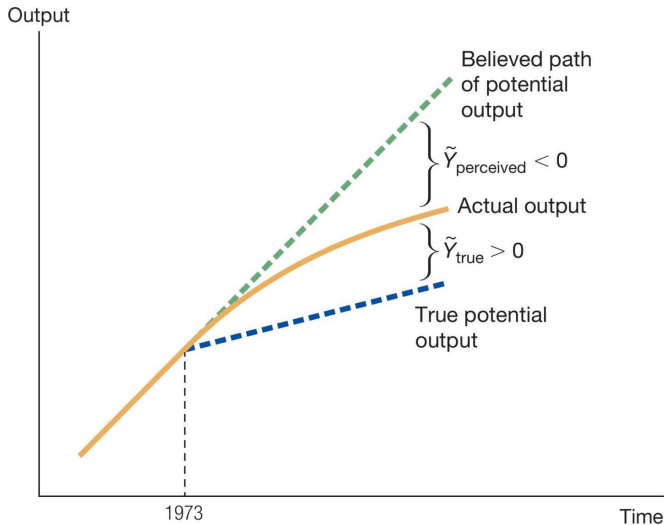


Figure 12.13 in Jones

Real time vs retrospective estimates of the output gap:

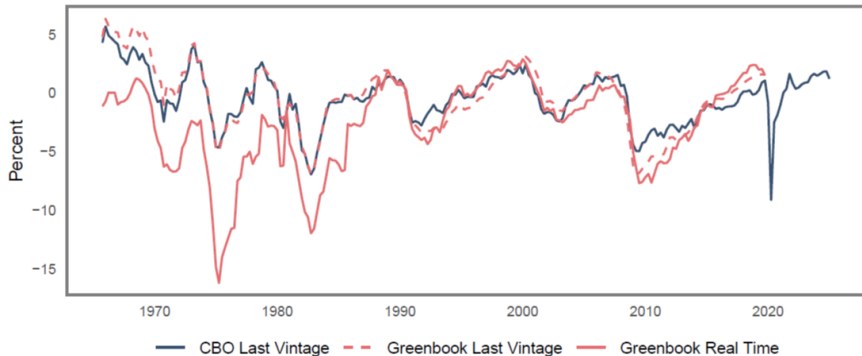
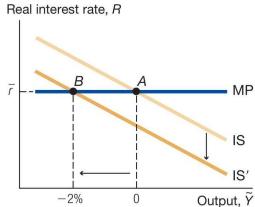


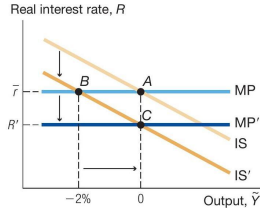
Figure 5: Estimates of the Output Gap in the U.S.

From Nakamura et al. (2025)

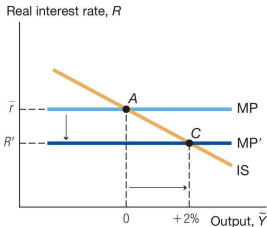
The great inflation of the 1970s



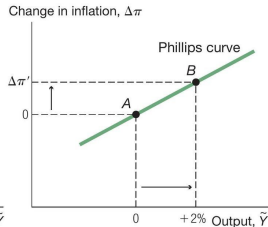
(a) Policymakers believe the slow-down in growth is caused by a negative aggregate demand shock...



(b) so they lower R .



(c) In truth, the IS curve never shifted...



(d) so $\downarrow R \Rightarrow \uparrow \tilde{Y} \Rightarrow \uparrow \pi$.